

Motivation and scale separation

- Thermal quantum field theory describes relativistic matter in equilibrium through imaginary time and Matsubara modes. In gauge plasmas, hard charged particles generate collective screening encoded by the hard-thermal-loop (HTL) effective theory [2–6].
- The same framework connects the electron–positron plasma of the early Universe with dense, comparatively cold matter relevant to compact stars.
- Weak coupling separates **hard** momenta $K \sim Q_{\text{hard}} \sim \max(\pi T, |\mu|)$ from **soft** photons $K \sim m_E \sim eQ_{\text{hard}}$.
- At next-to-next-to-next-to-leading order (N3LO), the overlap gives the **mixed** sector: a soft HTL-resummed photon dressed by hard fermion loops. QED has no separate purely soft sector because photons do not self-interact.
- In this work, we obtain the pressure contribution in the mixed sector at N3LO.

Quantity computed. After subtracting the overlap with the hard expansion, we evaluate [1]

$$p_{\text{mix}} = \frac{T}{2} \sum_{n=-\infty}^{\infty} \int \frac{d^d k}{(2\pi)^d} \sum_I g_I \frac{\mathcal{N}_I(K)}{K^2 [K^2 + \Pi_I^{\text{LO}}(K)]},$$

$$\mathcal{N}_I(K) = \Pi_I^{\text{LO}}(K) [\Pi_I^{\text{NLO}}(K) - \Pi_I^{\text{Pow}}(K) \Pi_I^{\text{LO}}(K)].$$

Here $I \in \{\text{T}, \text{L}\}$ labels transverse and longitudinal photons, while LO, NLO, and Pow denote the HTL, two-loop, and power-correction pieces of the self-energies. The metric is Euclidean, since we are in imaginary time with $K = (2\pi nT, \mathbf{k})$.

Static and nonstatic Matsubara modes

Static mode. The zero mode is analytic and begins at $\mathcal{O}(e^5)$:

$$p_{\text{mix}}^{(0)} = -\frac{e^2 T m_E^3}{\pi^3} \left[\frac{3}{64} + \frac{N_f}{96} (\mathcal{I}_T - 3) \right].$$

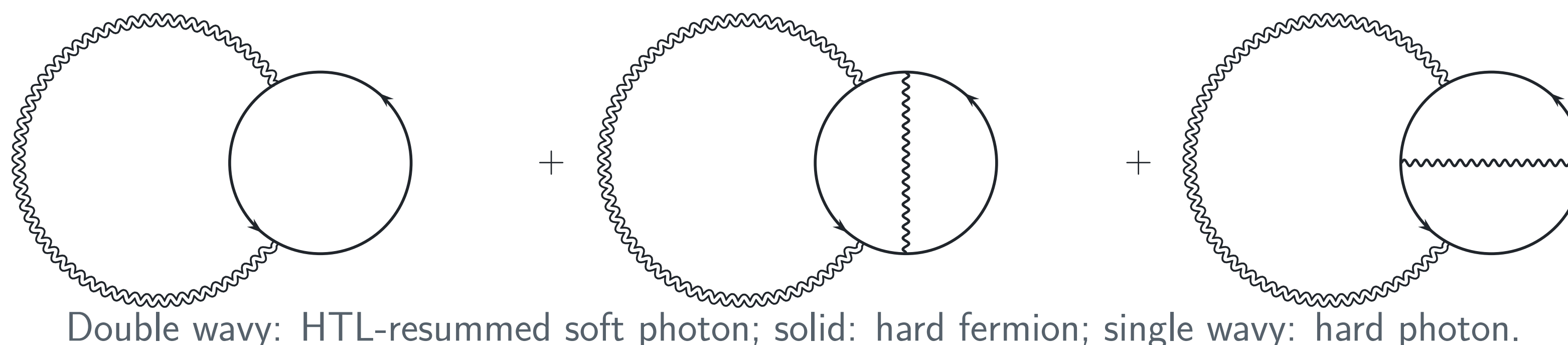
$\mathcal{I}_T$ is a function of T, μ , and the renormalisation scale $\bar{\Lambda}$. **Nonstatic modes.** For $n \neq 0$, an angular change of variables reduces the sum-integral to one numerical integral:

$$p_{\text{mix}}^{(n \neq 0)} = \int_0^{\pi/2} d\phi \sum_I w_I(\phi) \mathcal{F}_d(\alpha_I(\phi)),$$

$$\mathcal{F}_d(\alpha) = \frac{1}{2\epsilon} - \text{Re} \psi(1 + i\alpha) + \mathcal{O}(\epsilon).$$

Here $\alpha_I^2(\phi) = \Pi_I^{\text{LO}}(\phi) \cos^2 \phi / (2\pi T)^2$, while w_I contains the remaining loop and angular functions.

Mixed-sector topologies.



Pole structure. Dimensional regularization separates the mixed pressure as

$$p_{\text{mix}} = \frac{A_{-1}}{\epsilon} + A_0 + \mathcal{O}(\epsilon).$$

The pole is analytic and cancels against the infrared pole of the hard sector; only the one-dimensional digamma integral in A_0 remains numerical.

Limits, interpretation, and outlook

Thermal and cold limits.

- For $\alpha_I \rightarrow 0$, the high-temperature expansion is analytic in e^2 .
- For $\alpha_I \rightarrow \infty$, the cold limit generates an additional $e^6 \ln e$ contribution.

The finite part of $\mathcal{F}_d$ interpolates between

$$\mathcal{F}(\alpha) = \begin{cases} \gamma_E - \zeta(3)\alpha^2 + \mathcal{O}(\alpha^4), & \alpha \ll 1, \\ -\ln \alpha - \frac{1}{12\alpha^2} + \mathcal{O}(\alpha^{-4}), & \alpha \gg 1. \end{cases}$$

The large- α expansion reproduces the known zero-temperature result [7].

Future directions.

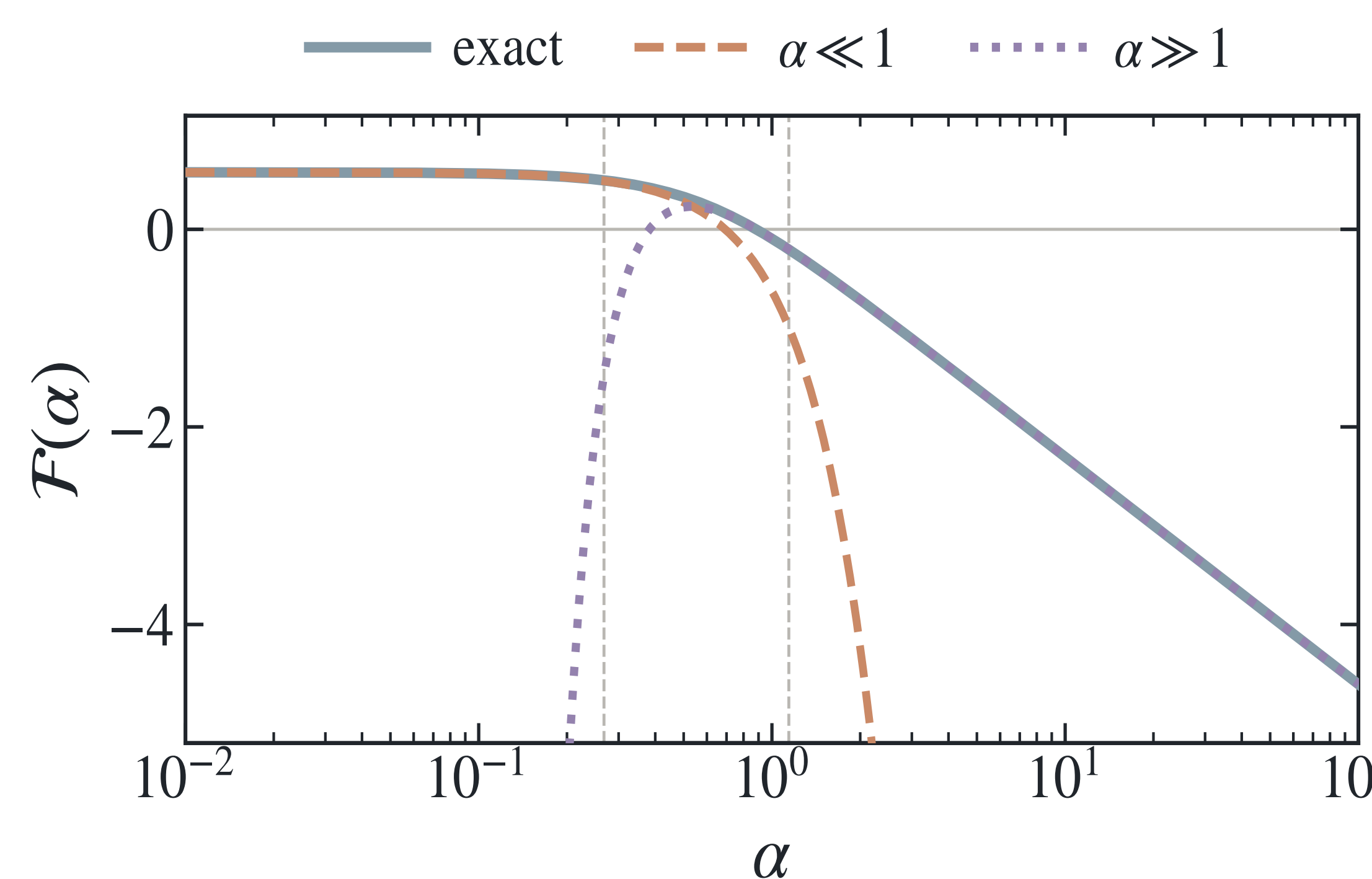
- Evaluate the remaining finite hard contribution and assemble the complete N3LO equation of state at arbitrary T/μ , including its thermodynamic derivatives and limiting regimes.
- Extend the finite- T , finite- μ framework to QCD, where non-Abelian soft contributions introduce additional scales and resummations relevant to hot and dense strongly interacting matter.

References

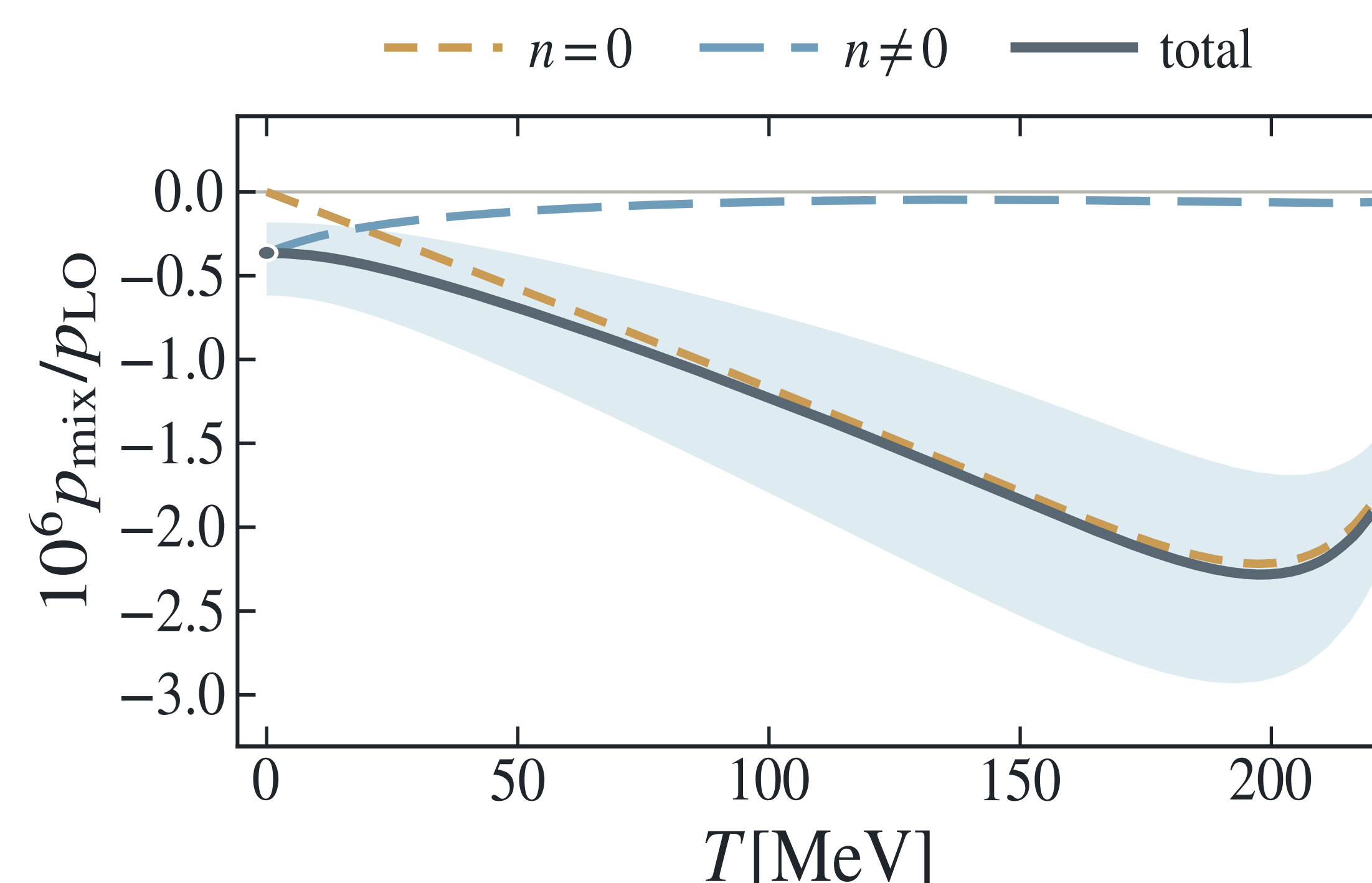
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Hot-to-cold crossover

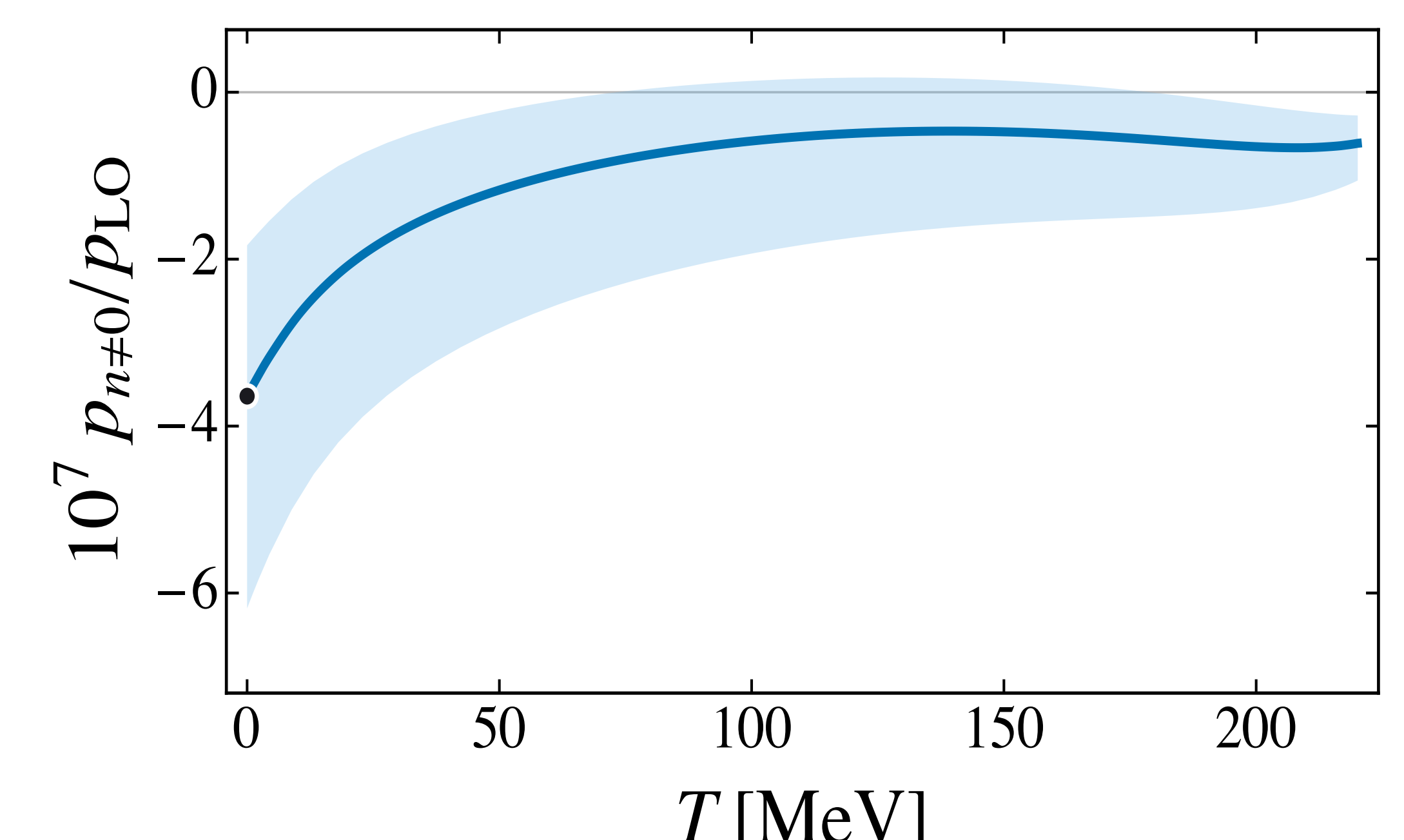
For $N_f = 1$, all pressure panels follow $\sqrt{(2\mu)^2 + (0.723 4\pi T)^2} = 2 \text{ GeV}$. The $X \in [1/2, 2]$ band varies $\bar{\Lambda} = \Lambda_h = X(2 \text{ GeV})$.



Finite Matsubara function's interpolation



Total mixed pressure versus T .



Nonzero-mode pressure versus T .